Friedrich-Alexander-Universität
Naturwissenschaftliche Fakultät

Bachelor QC II

Predicting UVvis absorption properties

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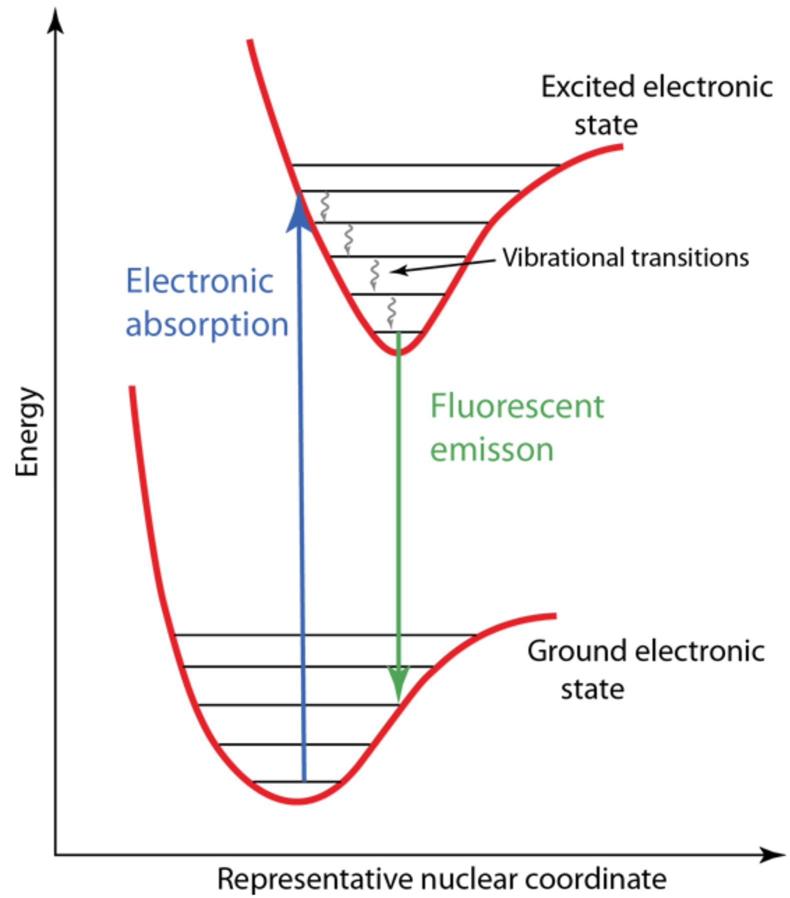
Stationary absorption and emission properties



Time dependent Perturbation Theory

time-dependent Schrödinger equation

$$-\frac{\hbar}{i}\frac{\partial}{\partial t}\Psi = \hat{H}\Psi, \quad \text{with } \Psi_j = \exp\left(-\frac{iE_j t}{\hbar}\right)\phi_j \tag{1}$$

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Let's perturb the Hamiltonian with a radiation field (electronic field):

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Wavefunction evolves in the presence of the perturbation and may be expressed as the linear combination of the complete set of solutions to $\hat{H}$:

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Termination of the radiation field will cause the WF to collapse to a stationary state with probability $|c_k|^2$. Let's put things together: (3), (2) $\rightarrow$ (1)

$$-\frac{\hbar}{i}\frac{\partial}{\partial t}\sum_k c_k \exp\left(-\frac{iE_k t}{\hbar}\right)\phi_k = \left[\hat{H}^0 + e_0\mathbf{r}\sin(2\pi\nu t)\right]\sum_k c_k \exp\left(-\frac{iE_k t}{\hbar}\right)\phi_k \tag{4}$$

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$$-\frac{\hbar}{i}\frac{\partial}{\partial t}\sum_k c_k \exp\left(-\frac{iE_kt}{\hbar}\right)\phi_k = \left[\hat{H}^0 + e_0\mathbf{r}\sin\left(2\pi\nu t\right)\right]\sum_k c_k \exp\left(-\frac{iE_kt}{\hbar}\right)\phi_k \qquad (5)$$

$$-\frac{\hbar}{i}\sum_k \frac{\partial c_k}{\partial t}\exp\left(-\frac{iE_kt}{\hbar}\right)\phi_k + \sum_k c_kE_k \exp\left(-\frac{iE_kt}{\hbar}\right)\phi_k =$$
$$\sum_k c_kE_k \exp\left(-\frac{iE_kt}{\hbar}\right)\phi_k + e_0\mathbf{r}\sin\left(2\pi\nu t\right)\sum_k c_k \exp\left(-\frac{iE_kt}{\hbar}\right)\phi_k$$

Time dependent Perturbation Theory

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$$-\frac{\hbar}{i}\sum_k \frac{\partial c_k}{\partial t}\exp\left(-\frac{iE_k t}{\hbar}\right)\phi_k = e_0\mathbf{r}\sin\left(2\pi\nu t\right)\sum_k c_k \exp\left(-\frac{iE_k t}{\hbar}\right)\phi_k \tag{6}$$

The change of c_k with time is dependent of the perturbing radiation field.

Time dependent Perturbation Theory

$$-\frac{\hbar}{i}\frac{\partial}{\partial t}\sum_k c_k \exp\left(-\frac{iE_k t}{\hbar}\right)\phi_k = \left[\hat{H}^0 + e_0\mathbf{r}\sin\left(2\pi\nu t\right)\right]\sum_k c_k \exp\left(-\frac{iE_k t}{\hbar}\right)\phi_k \qquad (5)$$

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$\langle\phi_m| \rightarrow$ (6), and integration

$$-\frac{\hbar}{i}\sum_k \frac{\partial c_k}{\partial t}\exp\left(-\frac{iE_k t}{\hbar}\right)\delta_{mk} = e_0\sin\left(2\pi\nu t\right)\sum_k c_k \exp\left(-\frac{iE_k t}{\hbar}\right)\langle\phi_m|\mathbf{r}|\phi_k\rangle \qquad (7)$$

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Assume a small perturbation, so we can use coefficients for the ground state ϕ_0:

$$\frac{\partial c_m}{\partial t} = -\frac{i}{\hbar}e_0 \sin\left(2\pi\nu t\right)\exp\left[-\frac{i\left(E_m - E_0\right)t}{\hbar}\right]\langle\phi_m|\mathbf{r}|\phi_0\rangle \tag{9}$$

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Integration of (9) over time gives:

$$c(\tau) = \frac{1}{2i\hbar}e_0\left[\frac{\exp\left(i(\omega_{m0}+\omega)\tau\right)-1}{\omega_{m0}+\omega} - \frac{\exp\left(i(\omega_{m0}-\omega)\tau\right)-1}{\omega_{m0}-\omega}\right]\langle\phi_m|\mathbf{r}|\phi_0\rangle$$

$$\omega_{m0} = \frac{E_m - E_0}{\hbar}; \quad \omega = 2\pi\nu$$

Time dependent Perturbation Theory

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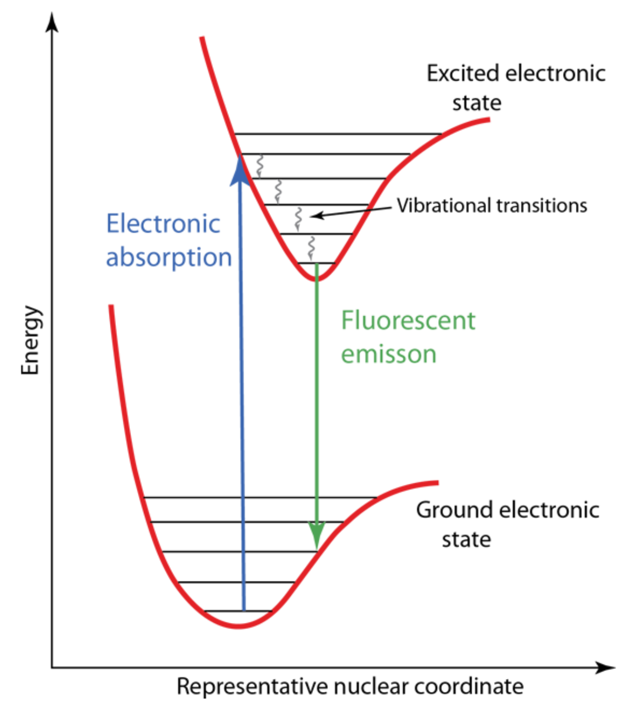
Qualitative points:

- $\langle \phi_m | \mathbf{r} | \phi_0 \rangle$:
- $\omega_{m0} - \omega$ in the denominator:



Time-dependent Density Functional Theory

Similar formalism as time-dependent perturbation theory.
$\rightarrow$ poles of polarizability matrix determine vertical excitation energies:

$$\langle \alpha \rangle_\omega = \sum_{m>0} \left[\frac{|\langle \phi_m | \mathbf{r} | \phi_0 \rangle|^2}{\omega_{m0} + \omega} - \frac{|\langle \phi_m | \mathbf{r} | \phi_0 \rangle|^2}{\omega_{m0} - \omega} \right]$$

- sensitive to functional
- problematic with charge transfer transitions

Computation of absorption properties

TDDFT in Orca 6

orca6 input

```
!B3LYP def2TZVP # FUNCTIONAL BASIS
%TDDFT
  NROOTS  20 # number of states
END
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TD-DFT EXCITED STATES (SINGLETS)
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the weight of the individual excitations are printed if larger than 1.0e-02

STATE  1:  E=   0.094004 au      2.558 eV    20631.4 cm**-1 <S**2> =   0.000000 Mult 1
    46a ->  48a  :     0.988678
    46a ->  52a  :     0.010160

STATE  2:  E=   0.128128 au      3.487 eV    28120.8 cm**-1 <S**2> =   0.000000 Mult 1
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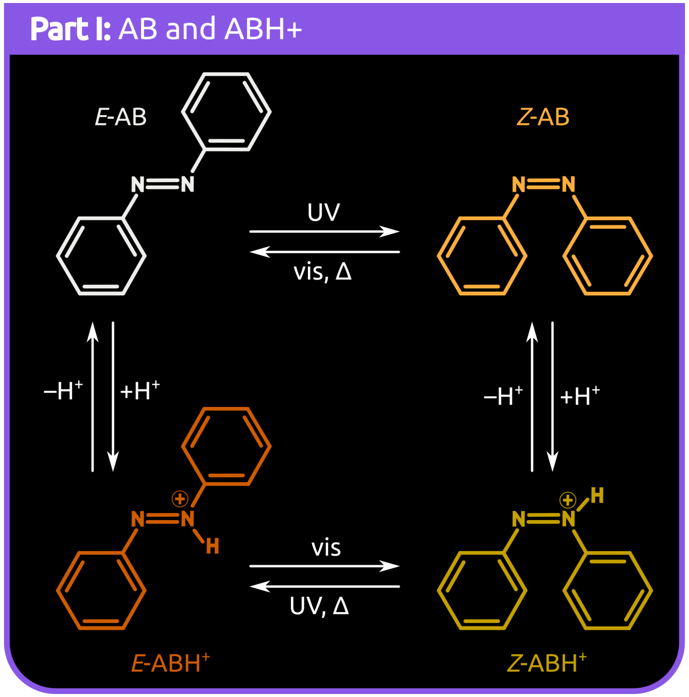
```
----------------------------------------------------------------------------------------------
                     ABSORPTION SPECTRUM VIA TRANSITION ELECTRIC DIPOLE MOMENTS
----------------------------------------------------------------------------------------------
     Transition         Energy     Energy   Wavelength fosc(D2)      D2        DX        DY        DZ
                         (eV)      (cm-1)     (nm)                 (au**2)    (au)      (au)      (au)
----------------------------------------------------------------------------------------------
  0-1A  ->  1-1A    2.557971   20631.4   484.7   0.000000000   0.00000  -0.00000   0.00000   0.00000
  0-1A  ->  2-1A    3.486530   28120.8   355.6   0.899300393  10.52818  -3.24007   0.17362  -0.00000
```

Computation of absorption data

pH dependent absorption spectra of azobenzenes

- Calculate absorption properties of E/Z-azobenzene derivatives
  - Excitation energies
  - Oscillator strengths
  - Character of states
- visualization and discussion of results
  - plot charge density differences
  - compare absorption properties of E/Z-isomers
  - compare absorption properties at different protonation stages



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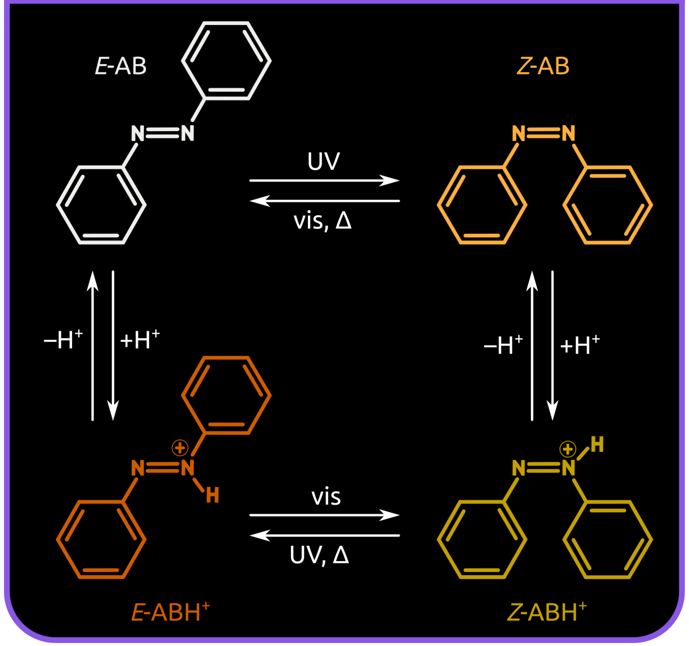
Part I: AB and ABH+



Instructions and sample inputs, coordinates etc.:

`https://github.com/CompPhotoChem/bachelor-qc-2/tree/main/azobenzene`